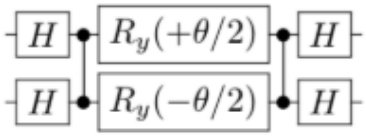
**Exploring Quantum Orthogonal Neural Networks**

The combination of quantum computing principles with neural network designs presents an exciting research and application frontier. Quantum orthogonal layers stand out as a pioneering concept in quantum machine learning. The paper 'Classical and Quantum Algorithms for Orthogonal Neural Networks' [1] introduces the Pyramidal Circuit, a new layer design that ensures orthogonality in neural network weights. This approach not only maintains orthogonality but does so more efficiently than classical computing methods. Orthogonality offers a significant advantage in neural networks to prevent vanishing or exploding gradients, thus potentially improving accuracy and stability in deep network training.

The architectures of quantum orthogonal neural networks are composed by Reconfigurable Beam Splitter (RBS) gates and specialized data loaders for encoding classical information into quantum states. An RBS gate is constructed using a combination of four Hadamard gates, two Ry rotation gates, and two CZ gates. In implementing these networks across multiple layers, a crucial step involves performing quantum tomography at each layer's end to extract the classical output. Following this, the output is re-encoded into the quantum domain after applying non-linear transformations, enabling the network to proceed to the next layer. This iterative process of tomography and re-encoding highlights the blend of quantum and classical techniques essential for operating orthogonal quantum neural networks efficiently.

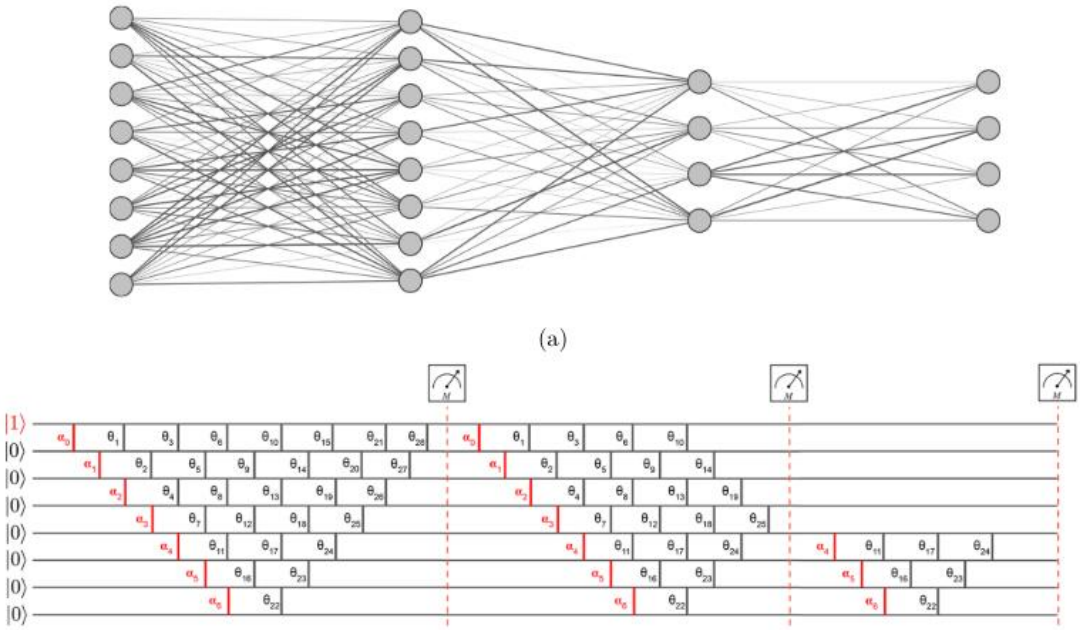



*A possible decomposition of the RBS(θ) gate.*

$$RBS(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

*RBS(θ) gate unitary matrix.*

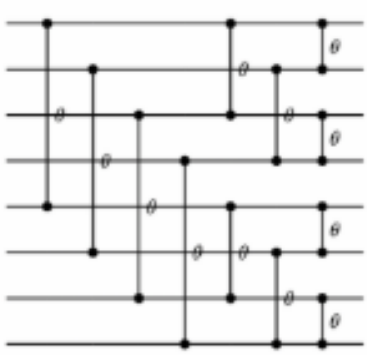
The Pyramidal Circuit highlights the efforts to use quantum computing's unique capabilities for enhancing machine learning. Quantum operations are intrinsically unitary (and thus orthogonal), a feature not easily replicated in classical computing without significant computational effort. By associating orthogonal matrix elements with quantum gate parameters, this paper proposes an efficient and orthogonality-preserving method for training neural networks.



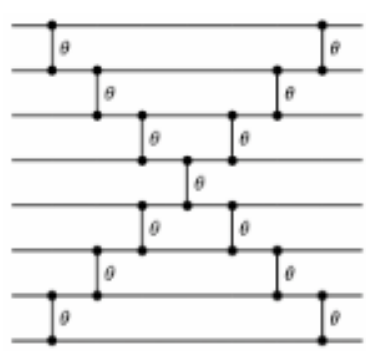

*Example of pyramidal circuits that build up a multilayer NN*

Building on these concepts, the paper 'Quantum Vision Transformers' [2] applies quantum neural networks to image recognition. This adaptation uses quantum orthogonal layers, showing quantum computing's potential in managing complex data. The Quantum Vision Transformer points to how quantum computing can improve classical machine learning models by enhancing computational efficiency and reducing model parameter needs without losing performance.

A notable advancement in the Quantum Vision Transformer research is the introduction of butterfly and X quantum orthogonal circuits. These circuits represent steps towards adapting transformer architectures for quantum computing. The butterfly circuit, while highly expressive, presents a challenge for current quantum hardware due to its need for all-to-all qubit connectivity. This requirement stretches beyond the capabilities of many existing quantum systems, illustrating a gap between theoretical advancements and practical implementation in quantum computing.



*Butterfly circuit*



*X circuit*

In closing, the exploration of quantum orthogonal layers and Quantum Vision Transformers underlines not just a promising avenue for scientific progress but a sight into a future where the intersection of quantum computing and neural network technologies shows unexplored capabilities. The advancements detailed in these studies are not mere theoretical exercises but stepping stones toward realizing a new paradigm in machine learning, where quantum computing doesn't just augment but significantly amplifies the potential of neural networks.

References

1. 'Classical and Quantum Algorithms for Orthogonal Neural Networks' Iordanis Kerenidis, Jonas Landman, and Natansh Mathur

2. 'Quantum Vision Transformers' El Amine Cherrat , Iordanis Kerenidis, Natansh Mathur, Jonas Landman, Martin Strahm , and Yun Yvonna Li